# Designing LowDrop Out Voltage (LDO) in Cadence.

By-
Jayakanth M V

# CONTENTS

1) Objective of the Project
- Aim & Specifications required
- Tools and softwares used
- Desired Block diagram

2)Approach
- BGR
- Folded Cascode
- Combining BGR & Folded Cascode

3) Analysis of circuit :Without Load
- DC,AC,Transient,STB Analysis
- PVT variations with SS,TT,FF,FS,SF : V:1.62 to 1.92,Temp:-40 to 125
- Undershoot with 6m Impulse
- Settling Time

4) Analysis of circuit :With Full Load
- DC,AC,Transient,STB Analysis
- PVT variations with SS,TT,FF,FS,SF : V:1.62 to 1.92,Temp:-40 to 125

5) Monte Carlo Simulations for Output with and without Load

6)Conclusion

# Objective of the Project

## Aim:

Designing the LDO with BGR with low voltage and Folded cascode Op-Amp with Pass transistor . All these blocks will be integrated for new LDO

## Specifications:

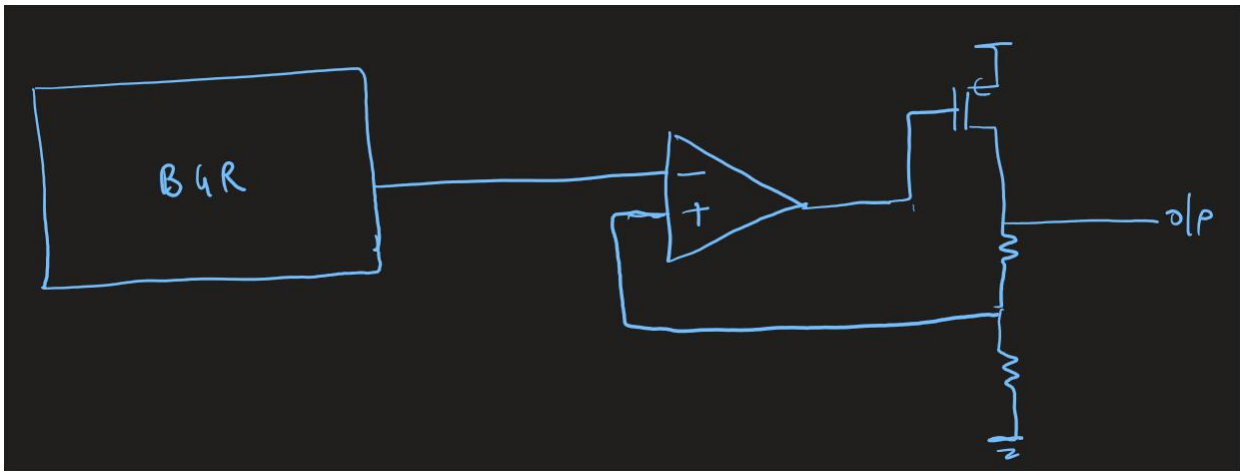
The specifications for BGR are as follows:

- Max load transient undershoot : 25mV for 6mA (give a sudden impulse of 6mA and optimize for undershoot)
- No-load settling time < 400ns.
- Supply Range : 1.8 +/- 10%
- Temperature Range : -40 ◌c to 125 ◌c
- Load capacitance : 60pF
- DC PSRR magnitude > 40dB, at other frequencies >10dB
- Phase margin at full-load >60deg, at no-load >45deg
- No-load power consumption < 500uW

## Tools & Softwares Used:

The Software used is Cadence Virtuoso and tools are ADE-XL , Maestro

## Desired Block Diagram:



# Approach

## BGR Design:

Band-gap Voltage Reference design is changed to Low voltage as per required specifications. To decrease the $V_{ref}$ we introduced same resistors $R_2$ at point where both has same voltage $V_{BE.}$ R is taken same as previous BGR

Resistor value i

Current in the BGR is $I_{REF}=V_T*\ln(N)/R + V_{BE}/R_2$

$$I_{REF}=(V_T*\ln(N)* R_2/R + V_{BE})/R_2$$

If we adjust $R_2/R$ ratio then $(V_T*\ln(N)* R_2/R + V_{BE}) = E_o\backslash q = 1.12v$

We have taken $R_2/R$ ratio as 1/10

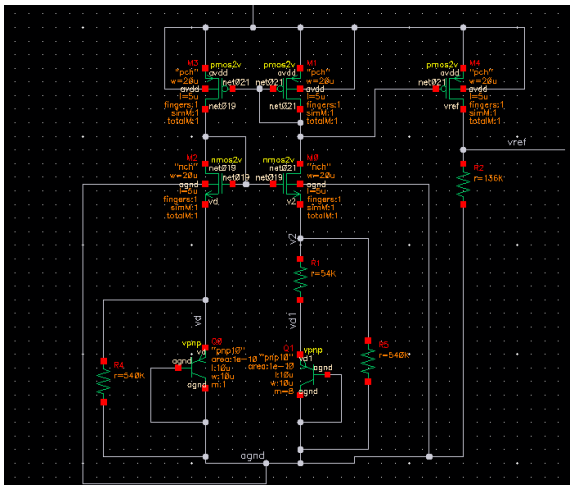
Finally at $R_3$ the output is $V_{ref} = (V_T*\ln(N)* R_2/R + V_{BE})/R_2*(R_3/ R_2)$

$$V_{ref} = 1.1*138k/540k$$

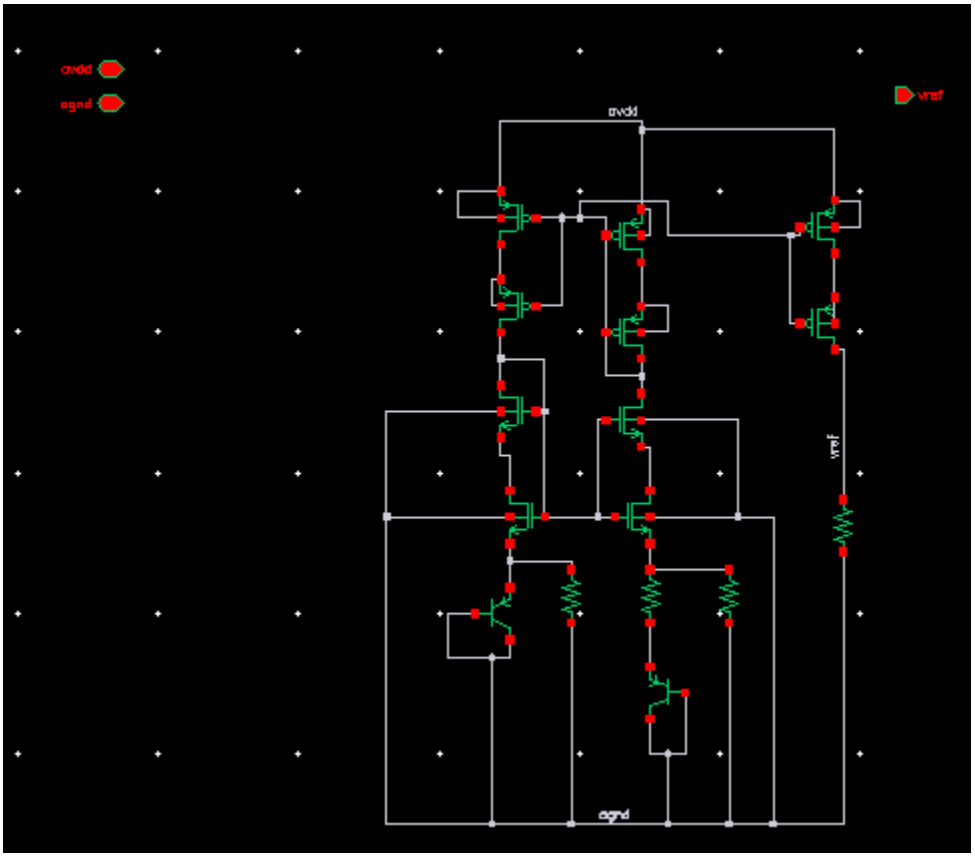
$$V_{ref} = 0.286$$

To increase PSRR every MOSFET in BGR is self cascoded which helps in impedance boosting of BGR

## Modified BGR Design:

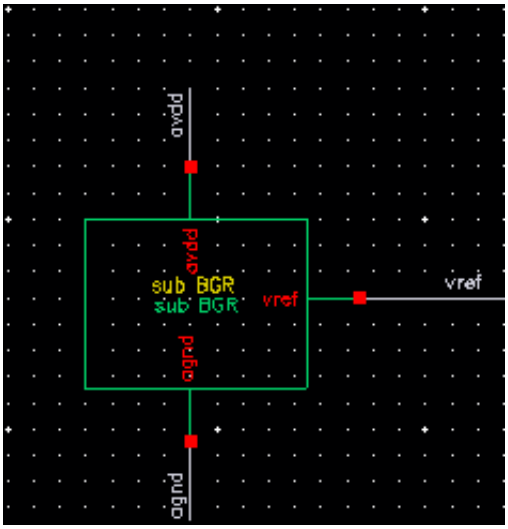


## Modified BGR Design to improve output impedance using Self Cascode:



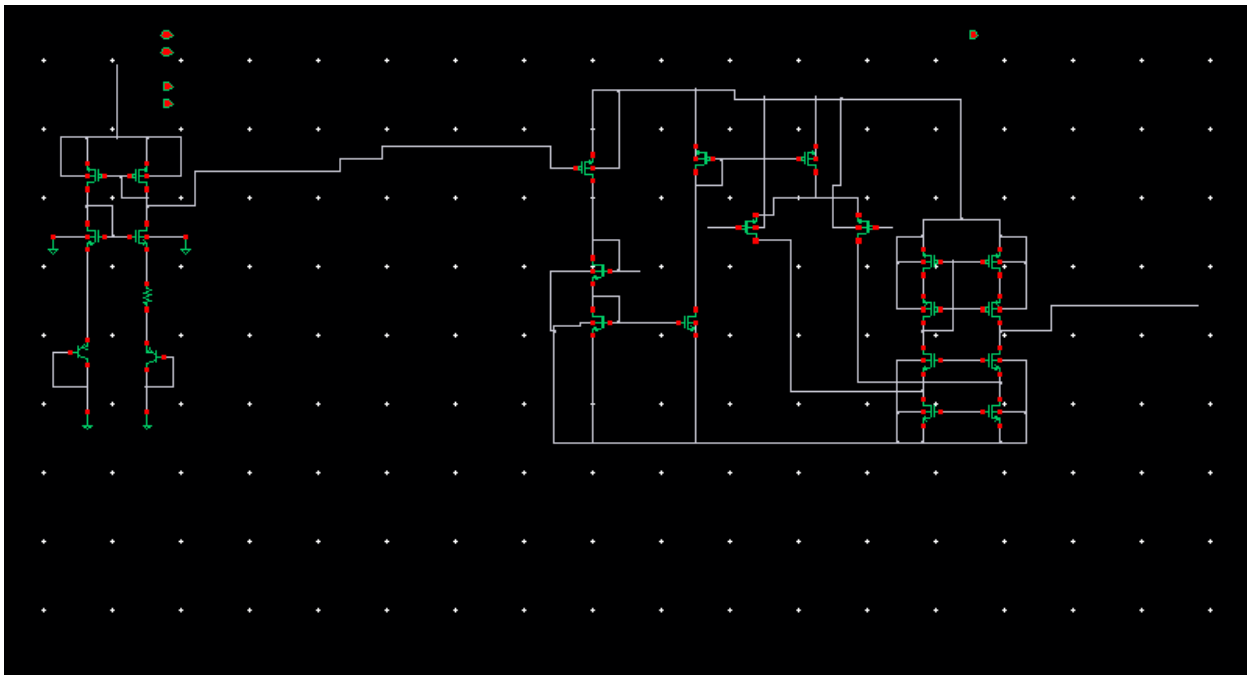

The output of BGR is made to get 0.3V ,it can be done by adjusting the 3rd branch Resistor

## Symbol:



## Folded Cascode:

With same specifications as mini project we used folded op amp but changed to PMOS because of Low Voltage coming out of BGR ie 0.3V



## Pass Transistor and Feedback Loop:

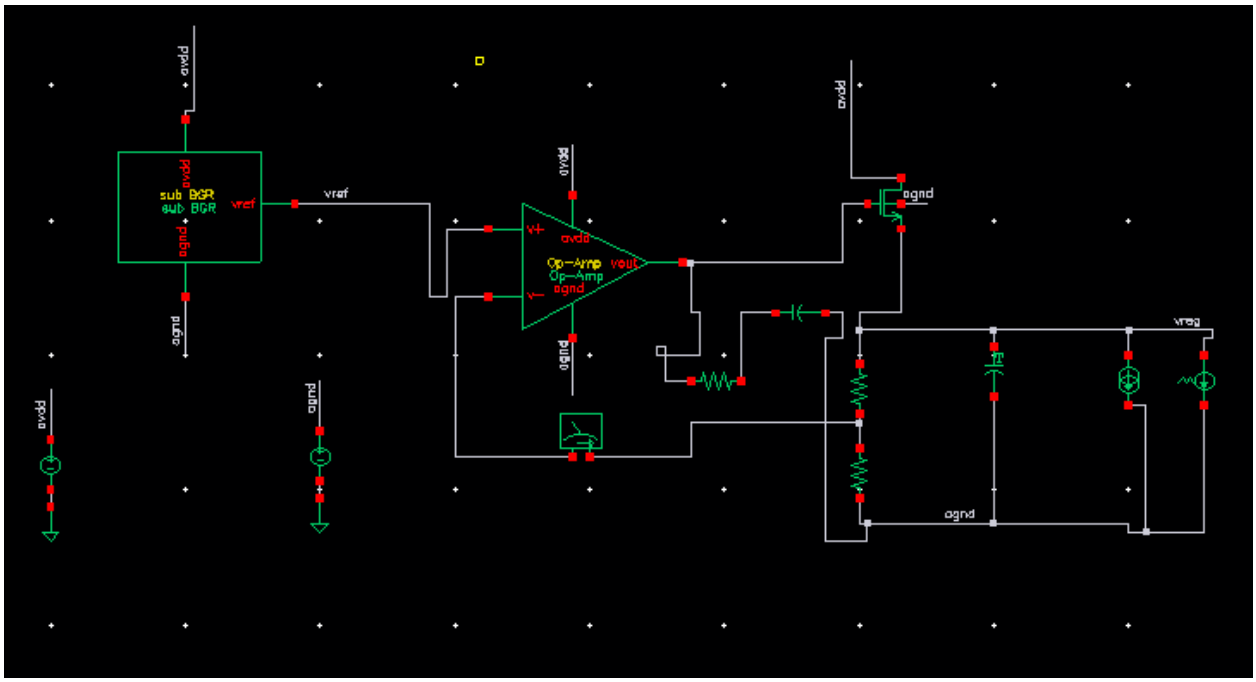
Design:

- Pass transistor is W/L ratio is taken such that it can carry current of 1.2mA as per by specifications.W/L=1/12*800
- Introduced miller pole compensation by adding capacitor and resistor in between output and OPAMP output to create a dominant pole at low frequency.
- Resistor values are taken such that it will provide output voltage 0.4V
- Considering Vout=0.4 and Vin as 0.3 from BGR

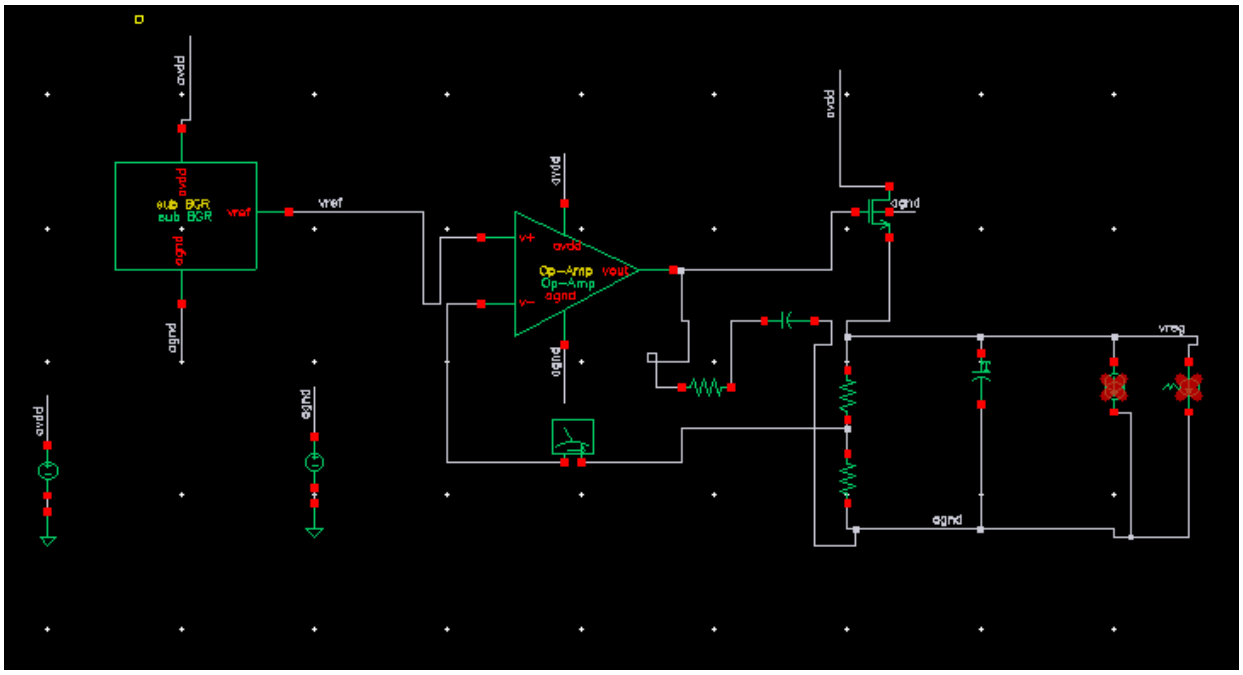
$$Vin=R_2/(R_1+R_2)*Vout$$

Gives ratio of $R_1/R_2=⅓$

## LDO Block Diagram:

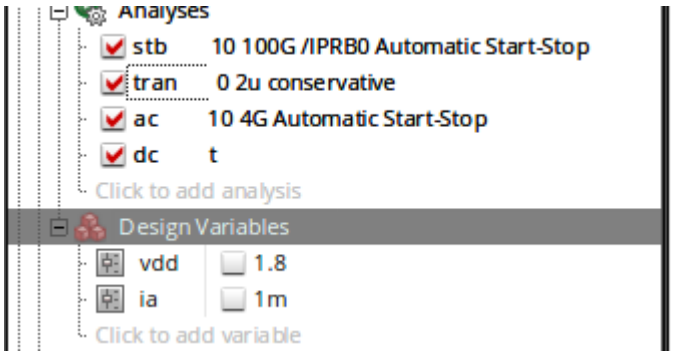



**Setup & Results**
**No Load Condition**



| Test | Name | Type | Details | EvalType | Plot | Save | Spec | Weight | Units | Digits | Notation | Suffix |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Final_proj:LDO_des:1 | IDC | expr | abs(IDC("/V1/PLUS")) | point | ☑ | ☐ | | | | | | |
| Final_proj:LDO_des:1 | VDD | expr | VDC("/avdd") | point | ☑ | ☐ | | | | | | |
| Final_proj:LDO_des:1 | POWER | expr | (VDD * IDC) | point | ☑ | ☐ | | | | | | |
| Final_proj:LDO_des:1 | | signal | /V1/PLUS | point | ☑ | ☑ | | | | | | |
| Final_proj:LDO_des:1 | PSRR | expr | dB20(VF("/vreg") / VF("/avdd")) | point | ☑ | ☐ | | | | | | |
| Final_proj:LDO_des:1 | | expr | VDC("/vref") | point | ☑ | ☐ | | | | | | |
| Final_proj:LDO_des:1 | op-amp output | expr | VDC("/net08") | point | ☑ | ☐ | | | | | | |
| Final_proj:LDO_des:1 | | expr | VT("/vreg") | point | ☐ | ☐ | | | | | | |
| Final_proj:LDO_des:1 | Phase Margin | expr | getData("phaseMargin" ?result "stb_margin") | point | ☑ | ☐ | | | | | | |
| Final_proj:LDO_des:1 | 3dB BW | expr | bandwidth(VF("/vreg") / VF("/net015")) 3 "lo... | point | ☑ | ☐ | | | | | | |
| Final_proj:LDO_des:1 | UGB | expr | bandwidth(VF("/vreg") / VF("/net015")) 0.1 "l... | point | ☑ | ☐ | | | | | | |
| Final_proj:LDO_des:1 | Gain | expr | dB20(VF("/vreg") / VF("/net015")) | point | ☑ | ☐ | | | | | | |
| Final_proj:LDO_des:1 | Phase | expr | phaseDegUnwrapped(VF("/vreg") / VF("/net0... | point | ☑ | ☐ | | | | | | |
| Final_proj:LDO_des:1 | Tranient Vreg | expr | VT("/vreg") | point | ☑ | ☐ | | | | | | |
| Final_proj:LDO_des:1 | Vreg | expr | VDC("/vreg") | point | ☑ | ☐ | | | | | | |
| Final_proj:LDO_des:1 | DC-PSRR | expr | value(dB20((VF("/vreg") / VF("/avdd"))) 0) | point | ☑ | ☐ | | | | | | |
| Final_proj:LDO_des:1 | MAX-PSRR | expr | ymax(dB20((VF("/vreg") / VF("/avdd")))) | point | ☑ | ☐ | | | | | | |

## Analysis

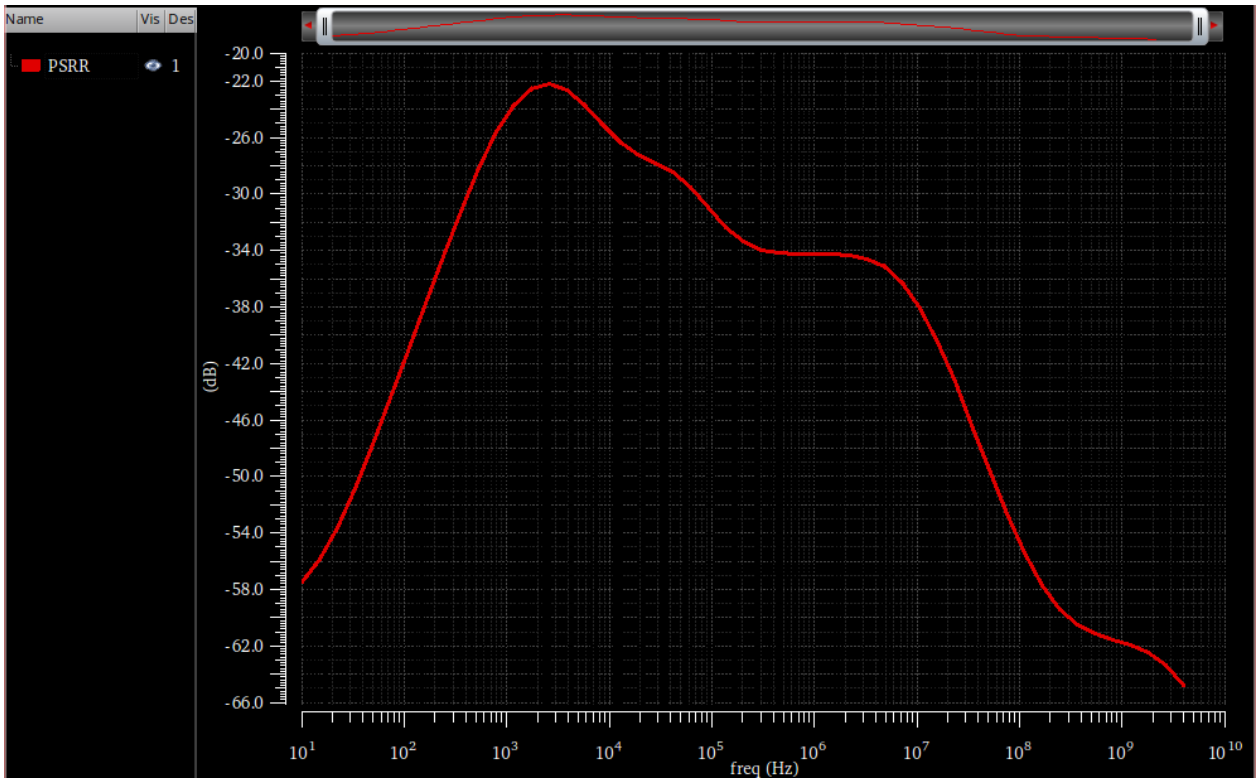



## Results

## No Load Condition: Nominal

| Test | Output | Nominal | Spec | Weight | Pass/Fail |
|---|---|---|---|---|---|
| Final_proj:LDO_des:1 | IDC | 107.2u | | | |
| Final_proj:LDO_des:1 | VDD | 1.8 | | | |
| Final_proj:LDO_des:1 | POWER | 193u | | | |
| Final_proj:LDO_des:1 | /V1/PLUS | | | | |
| Final_proj:LDO_des:1 | PSRR | | | | |
| Final_proj:LDO_des:1 | op-amp output | 941.8m | | | |
| Final_proj:LDO_des:1 | Phase Margin | 89.98 | | | |
| Final_proj:LDO_des:1 | 3dB BW | 55.93M | | | |
| Final_proj:LDO_des:1 | UGB | 8.584M | | | |
| Final_proj:LDO_des:1 | Gain | | | | |
| Final_proj:LDO_des:1 | Phase | | | | |
| Final_proj:LDO_des:1 | Tranient Vreg | | | | |
| Final_proj:LDO_des:1 | Vreg | 405.8m | | | |
| Final_proj:LDO_des:1 | DC-PSRR | -57.38 | | | |
| Final_proj:LDO_des:1 | MAX-PSRR | -19.98 | | | |
| Final_proj:LDO_des:1 | Gain_val | 41.39 | | | |

## PSRR:




Got DC value PSRR as -58, max is -22 ie everything is below -22

## Results PVT Variations

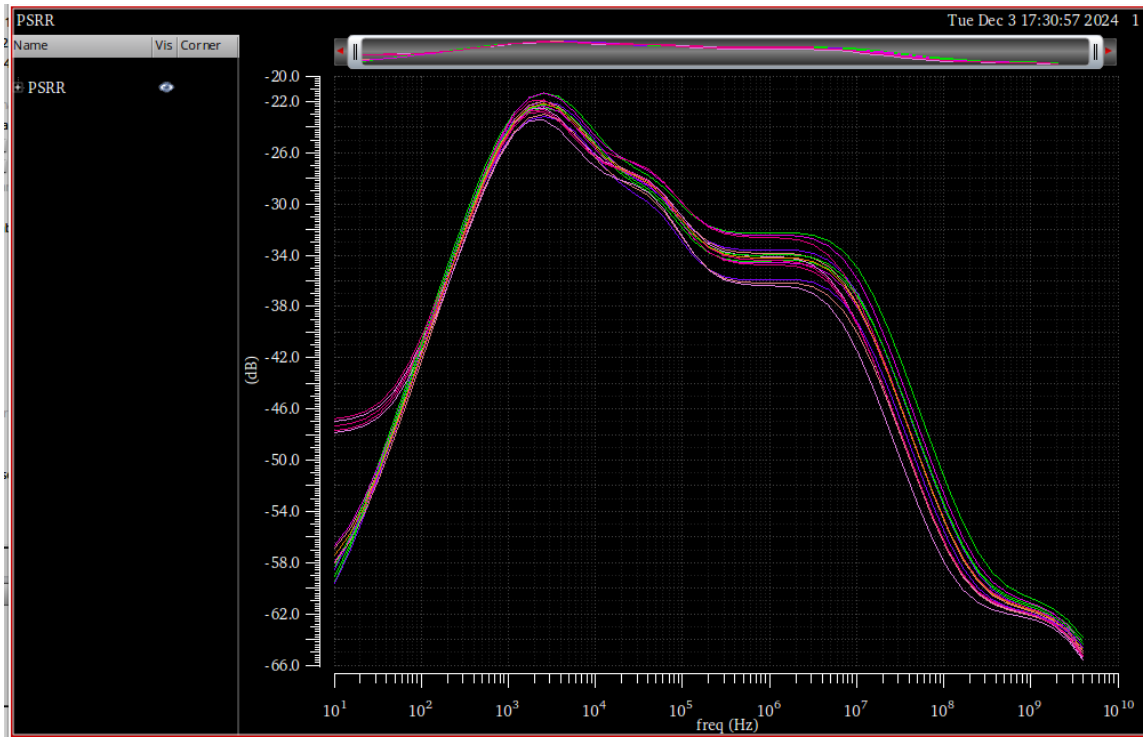
## No Load Condition

| Test | Output | Nominal | Spec | Weight | Pass/Fail | Min | Max | C0_0 | C0_1 | C0_2 | C0_3 | C0_4 | C0_5 | C0_6 | C0_7 | C0_8 | C0_9 | C0_10 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Final_proj:LDO_des:1 | IDC | 107.2u | | | | 100.3u | 112.1u | 111.8u | 107.1u | 100.3u | 111.9u | 107.2u | 100.5u | 112u | 107.3u | 100.8u | 111.9u | 107.1u |
| Final_proj:LDO_des:1 | VDD | 1.8 | | | | 1.62 | 1.98 | 1.62 | 1.62 | 1.62 | 1.8 | 1.8 | 1.8 | 1.98 | 1.98 | 1.98 | 1.62 | 1.62 |
| Final_proj:LDO_des:1 | POWER | 193u | | | | 162.5u | 221.9u | 181.1u | 173.5u | 162.5u | 201.4u | 193u | 181u | 221.7u | 212.5u | 199.6u | 181.2u | 173.5u |
| Final_proj:LDO_des:1 | /V1/PLUS | | | | | | | | | | | | | | | | | |
| Final_proj:LDO_des:1 | PSRR | | | | | | | | | | | | | | | | | |
| Final_proj:LDO_des:1 | op-amp output | 941.8m | | | | 760.5m | 1.087 | 1.027 | 946.5m | 832.2m | 1.022 | 941.8m | 828.1m | 1.018 | 937.3m | 824.4m | 1.087 | 1.008 |
| Final_proj:LDO_des:1 | Phase Margin | 89.98 | | | | 89.98 | 89.99 | 89.99 | 89.98 | 89.98 | 89.98 | 89.98 | 89.98 | 89.98 | 89.98 | 89.98 | 89.99 | 89.99 |
| Final_proj:LDO_des:1 | 3dB BW | 55.93M | | | | 41.9M | 71.31M | 60M | 56.32M | 51.77M | 59.73M | 55.93M | 51.15M | 59.56M | 55.74M | 50.87M | 48.67M | 45.85M |
| Final_proj:LDO_des:1 | UGB | 8.584M | | | | 6.524M | 10.91M | 9.175M | 8.638M | 7.969M | 9.132M | 8.584M | 7.892M | 9.095M | 8.553M | 7.844M | 7.686M | 7.3M |
| Final_proj:LDO_des:1 | Gain | | | | | | | | | | | | | | | | | |
| Final_proj:LDO_des:1 | Phase | | | | | | | | | | | | | | | | | |
| Final_proj:LDO_des:1 | Tranient Vreg | | | | | | | | | | | | | | | | | |
| Final_proj:LDO_des:1 | Vreg | 405.8m | | | | 375.6m | 427.1m | 426.3m | 405.6m | 375.7m | 426.4m | 405.8m | 376.3m | 426.6m | 406m | 377.2m | 426.6m | 405.7m |
| Final_proj:LDO_des:1 | DC-PSRR | -57.38 | | | | -59.83 | -44.47 | -58.58 | -58.03 | -49.42 | -59.03 | -57.38 | -47.27 | -59.25 | -56.09 | -45.03 | -59.09 | -58.56 |
| Final_proj:LDO_des:1 | MAX-PSRR | -19.98 | | | | -20.56 | -19.2 | -19.6 | -19.83 | -19.94 | -19.76 | -19.98 | -20.09 | -19.89 | -20.1 | -20.21 | -19.97 | -20.2 |
| Final_proj:LDO_des:1 | Gain_val | 41.39 | | | | 39.39 | 42.62 | 39.91 | 41.18 | 41.84 | 40.13 | 41.39 | 42.04 | 40.34 | 41.59 | 42.23 | 39.39 | 40.76 |

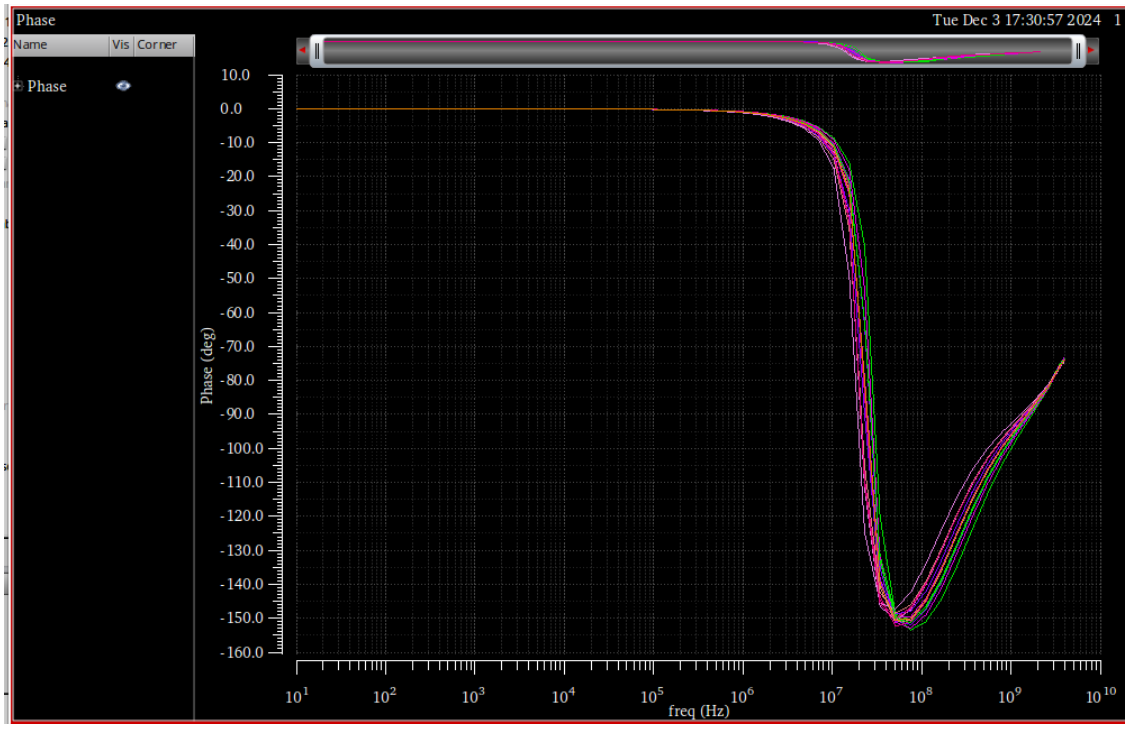
Results Obtained:

- Required Spec was less than +-10%: variation which was achieved
- UGB at Nominal 8M
- 3dB BW 55.93M
- Power Consumption less than 300u across all corners
- PSRR below 21 everywhere across all corners and PVT
- At DC PSRR is below -46 to -59

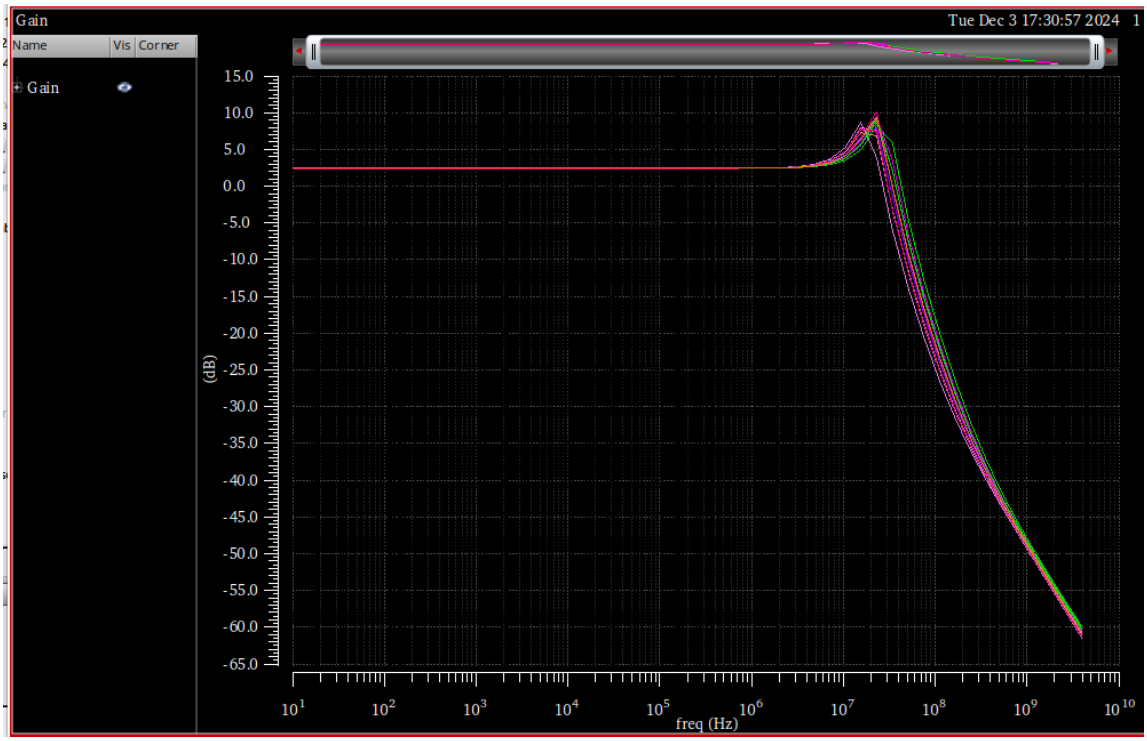
## PSRR:


PSRR
Tue Dec 3 17:30:57 2024
Name
Vis
Corner
PSRR
-20.0
-22.0
-26.0
-30.0
-34.0
-38.0
-42.0
-46.0
-50.0
-54.0
-58.0
-62.0
-66.0
(dB)
$10^1$ $10^2$ $10^3$ $10^4$ $10^5$ $10^6$ $10^7$ $10^8$ $10^9$ $10^{10}$
freq (Hz)

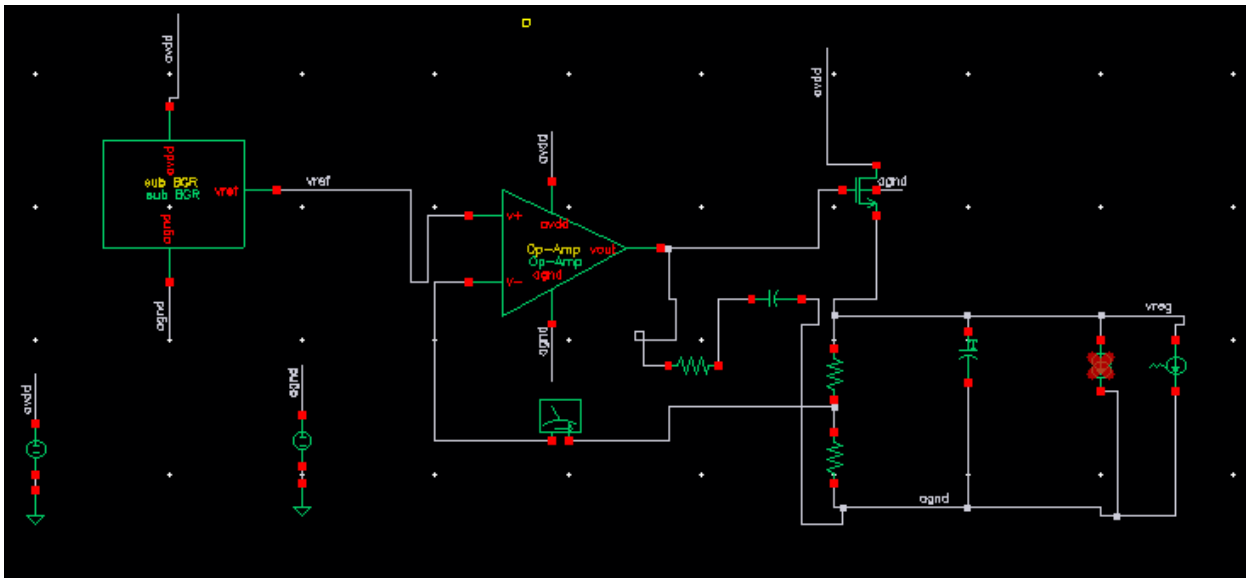
## Phase:

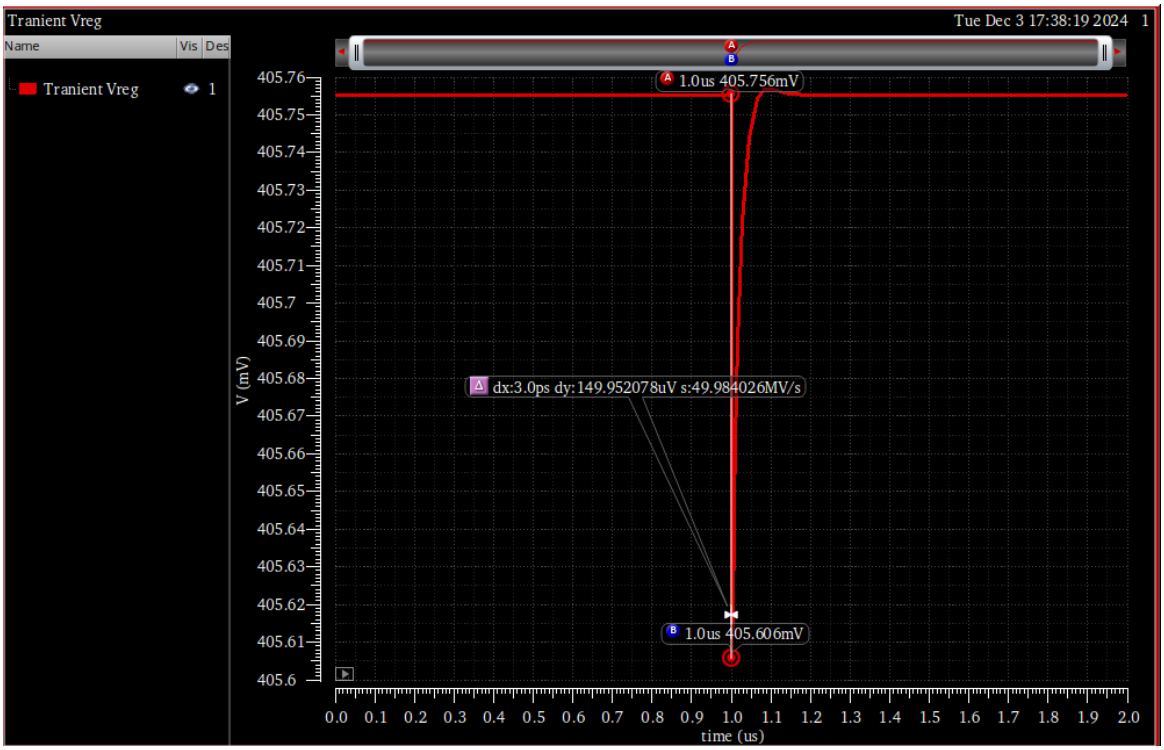

Phase
Tue Dec 3 17:30:57 2024
Name
Vis
Corner
Phase
10.0
0.0
-10.0
-20.0
-30.0
-40.0
-50.0
-60.0
-70.0
-80.0
-90.0
-100.0
-110.0
-120.0
-130.0
-140.0
-150.0
-160.0
Phase (deg)
$10^1$ $10^2$ $10^3$ $10^4$ $10^5$ $10^6$ $10^7$ $10^8$ $10^9$ $10^{10}$
freq (Hz)

## Gain:



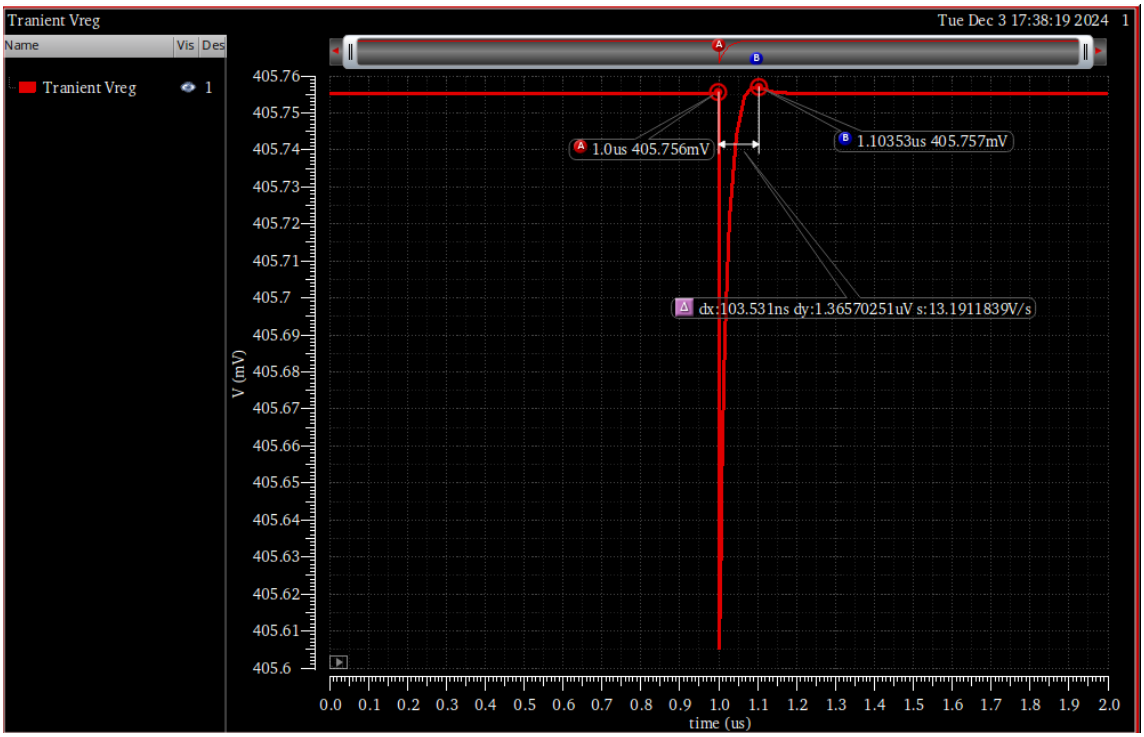

## MAX load undershoot:
## Giving impulse I_load=6mA





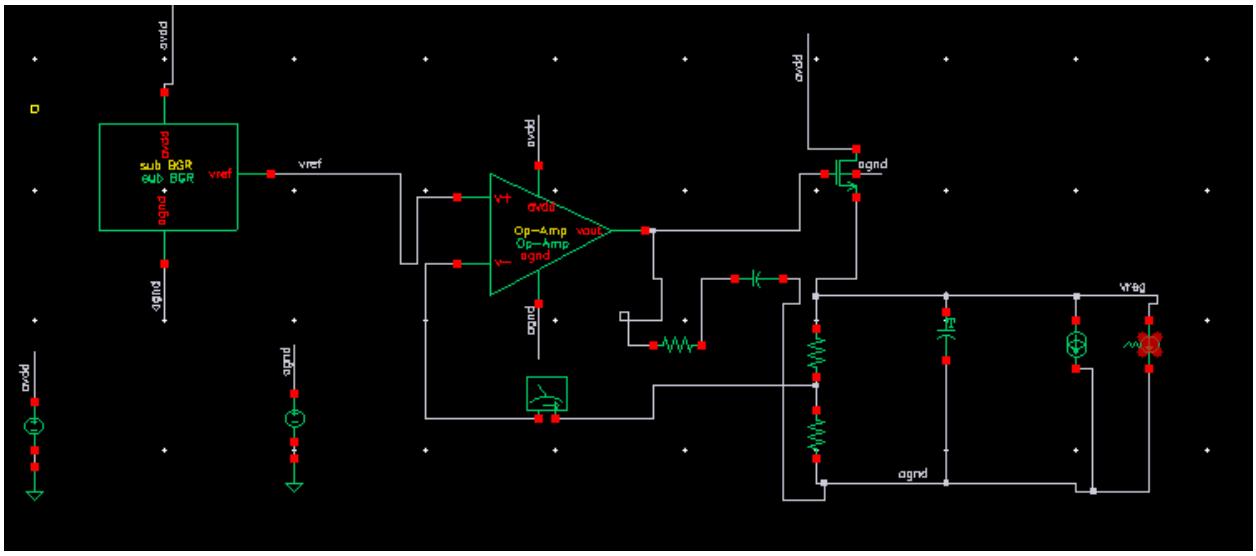

Got Max undershoot at 405.66mV , the undershoot value as 0.14mV

## Settling Time:




Got Settling Time as 103ns

# Setup & Results
# Full Load Condition

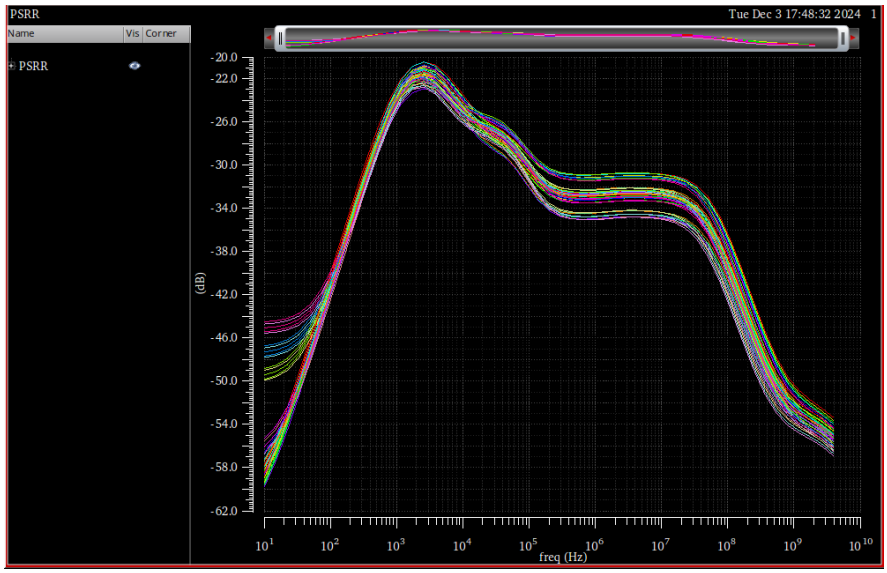



| | | |
|---|---|---|
| **Corners** | ☑ Nominal | ☑ C0 |
| | | |
| **Temperature** | | -40 27 125 |
| **Design Variables** | | |
| vdd | | 1.62 1.8 1.98 |
| Click to add | | |
| **Parameters** | | |
| Click to add | | |
| **Model Files** | | |
| cor_bip.scs | | ☑ tt_bip |
| cor_mim.scs | | ☑ tt_mim |
| cor_na.scs | | ☑ tt_na |
| cor_res.scs | | ☑ tt_res |
| cor_rfres_rpo.scs | | ☑ tt_rfres_rpo |
| cor_std_mos.scs | | ☑ tt ss ff sf fs |
| cr018gpii_v1d0.scs | | ☐ mc_mim |
| ...18gpii_v1d0.scs:1 | | ☑ stat_noise |
| Click to add | | |
| **Model Group(s)** | | <modelgroup> |
| Click to add | | |
| **Tests** | | |
| ☑ ..roj:LDO_des:1 | ☑ | ☑ |
| **...mber of Corners** | 1 | 45 |

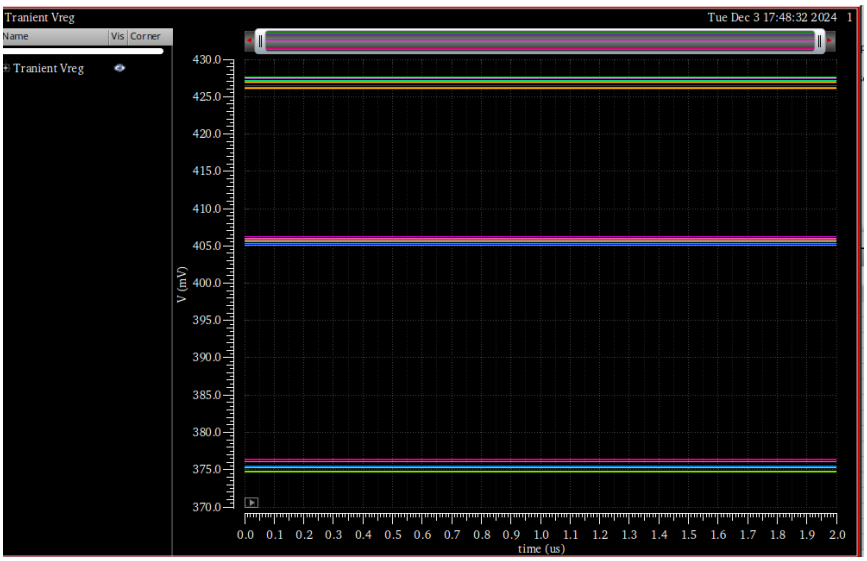
# Results PVT Variations

| Test | Output | Nominal | Spec | Weight | Pass/Fail | Min | Max | C0_0 | C0_1 | C0_2 | C0_3 | C0_4 | C0_5 | C0_6 | C0_7 | C0_8 | C |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Filter | Filter | Filter | Filter | Filter | Filter | Filter | Filter | Filter | Filter | Filter | Filter | Filter | Filter | Filter | Filter | Filter | Filter |
| Final_proj:LDO_des:1 | IDC | 1.107m | | | | 1.102m | 1.11m | 1.11m | 1.107m | 1.102m | 1.11m | 1.107m | 1.102m | 1.11m | 1.107m | 1.103m | 1.1 |
| Final_proj:LDO_des:1 | VDD | 1.8 | | | | 1.62 | 1.98 | 1.62 | 1.62 | 1.62 | 1.8 | 1.8 | 1.8 | 1.98 | 1.98 | 1.98 | 1 |
| Final_proj:LDO_des:1 | POWER | 1.993m | | | | 1.785m | 2.197m | 1.797m | 1.794m | 1.786m | 1.997m | 1.993m | 1.984m | 2.197m | 2.193m | 2.183m | 1.7 |
| Final_proj:LDO_des:1 | /V1/PLUS | | | | | | | | | | | | | | | | |
| Final_proj:LDO_des:1 | PSRR | | | | | | | | | | | | | | | | |
| Final_proj:LDO_des:1 | VDC("/vref") | 304.3m | | | | 281.6m | 320.3m | 319.7m | 304.2m | 281.7m | 319.8m | 304.3m | 282.1m | 319.9m | 304.4m | 282.8m | 31 |
| Final_proj:LDO_des:1 | op-amp output | 1.07 | | | | 901.9m | 1.212 | 1.143 | 1.075 | 986.4m | 1.138 | 1.07 | 981.8m | 1.133 | 1.065 | 977.7m | 1. |
| Final_proj:LDO_des:1 | Phase Margin | 89.98 | | | | 89.97 | 89.99 | 89.98 | 89.98 | 89.98 | 89.98 | 89.98 | 89.98 | 89.98 | 89.98 | 89.97 | 8 |
| Final_proj:LDO_des:1 | 3dB BW | 86.41M | | | | 68.89M | 102.2M | 90.13M | 87.53M | 82.88M | 89.25M | 86.41M | 81.34M | 88.6M | 85.66M | 80.28M | 74. |
| Final_proj:LDO_des:1 | UGB | 66.37M | | | | 55.33M | 83.17M | 68.17M | 67.31M | 65.89M | 67.28M | 66.37M | 64.93M | 66.67M | 65.77M | 64.3M | 57. |
| Final_proj:LDO_des:1 | Gain | | | | | | | | | | | | | | | | |
| Final_proj:LDO_des:1 | Phase | | | | | | | | | | | | | | | | |
| Final_proj:LDO_des:1 | Tranient Vreg | | | | | | | | | | | | | | | | |
| Final_proj:LDO_des:1 | Vreg | 405.7m | | | | 374.7m | 427.7m | 426.9m | 405.6m | 374.8m | 427m | 405.7m | 375.5m | 427.2m | 405.9m | 376.3m | 42 |
| Final_proj:LDO_des:1 | DC-PSRR | -57.26 | | | | -59.71 | -44.47 | -58.39 | -57.87 | -49.38 | -58.88 | -57.26 | -47.26 | -59.12 | -56.01 | -45.03 | -5 |
| Final_proj:LDO_des:1 | MAX-PSRR | -21.47 | | | | -22.89 | -20.42 | -21.31 | -21.16 | -21.66 | -21.62 | -21.47 | -21.83 | -21.95 | -21.67 | -21.96 | -2 |

# PSRR:




# Transient Response: Output wrt Time

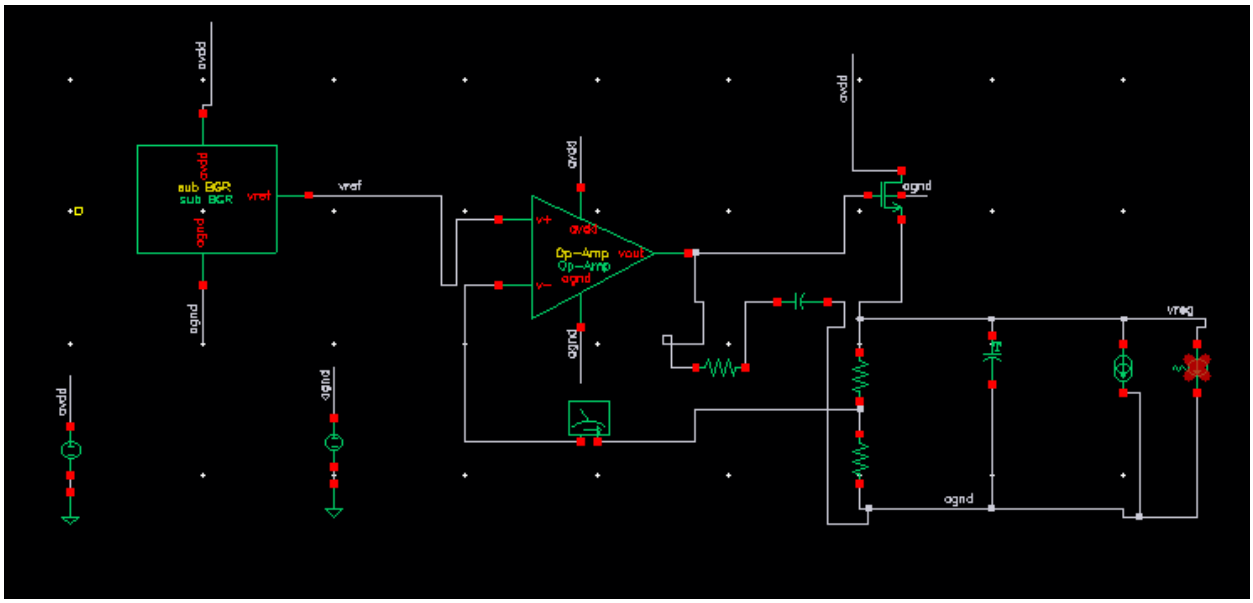


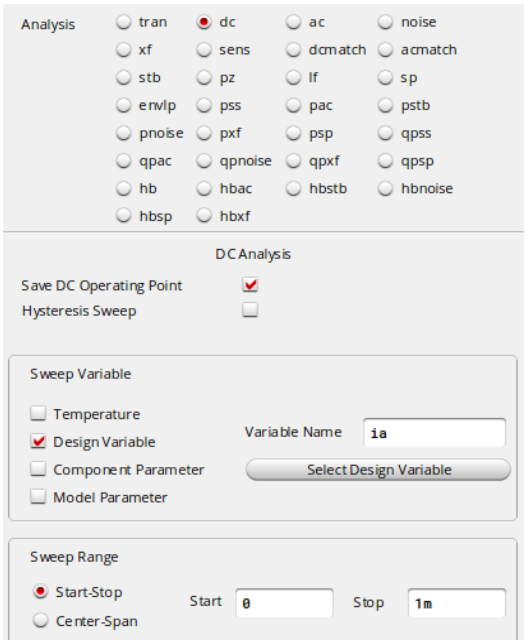
Results Obtained:

- Less than +-10%: variation which was achieved
- UGB at Nominal 66.37M, rest all 55.33M
- 3dB BW 86M at Nominal ,rest all above 68.89M
- PSRR below 20 everywhere across all corners and PVT
- At DC PSRR is below -44 to -58

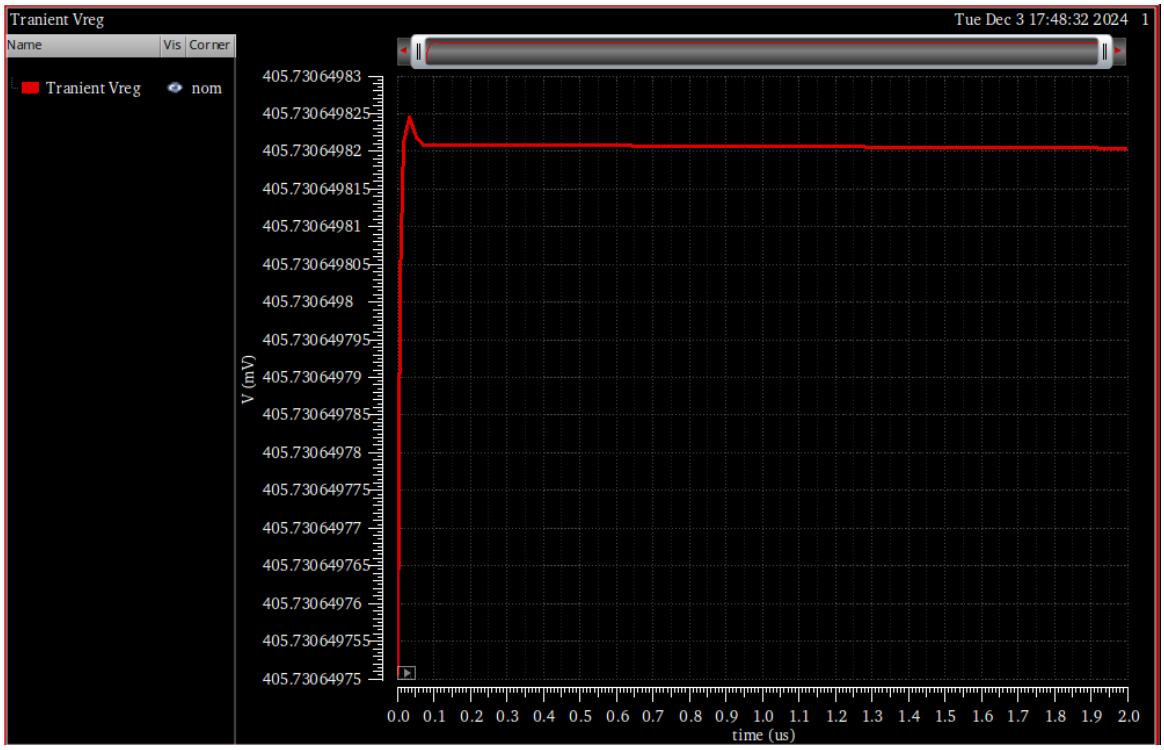
## Setup & Results

## Varying load and output:

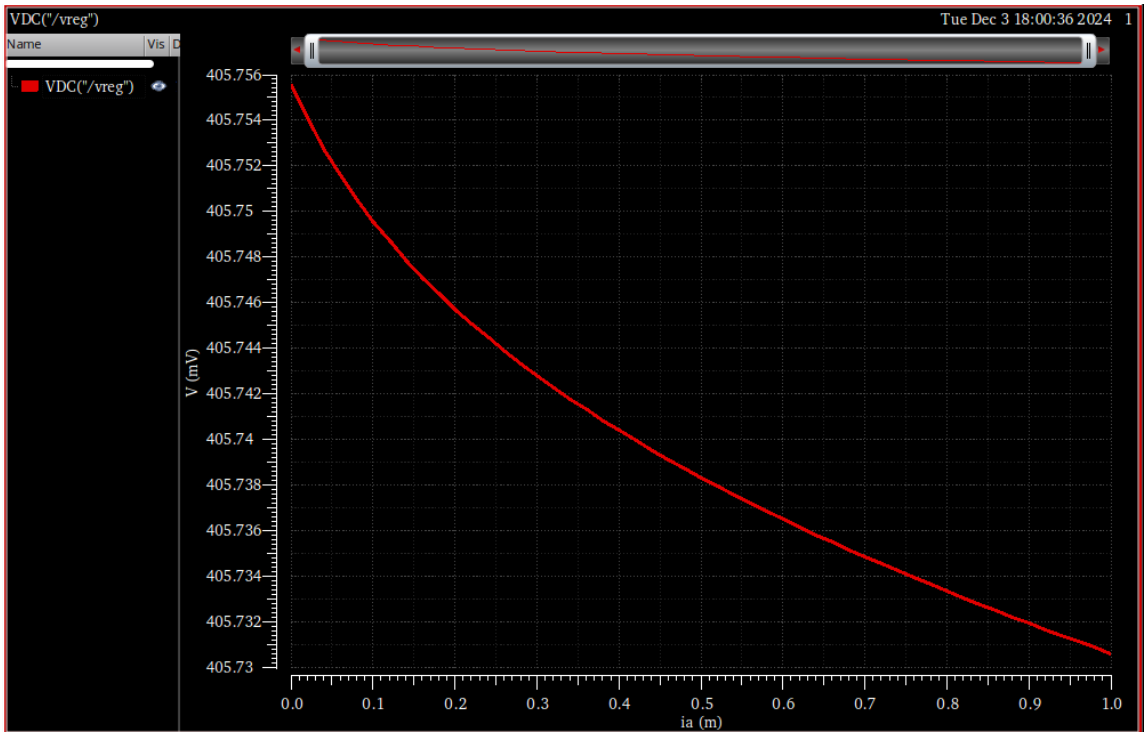


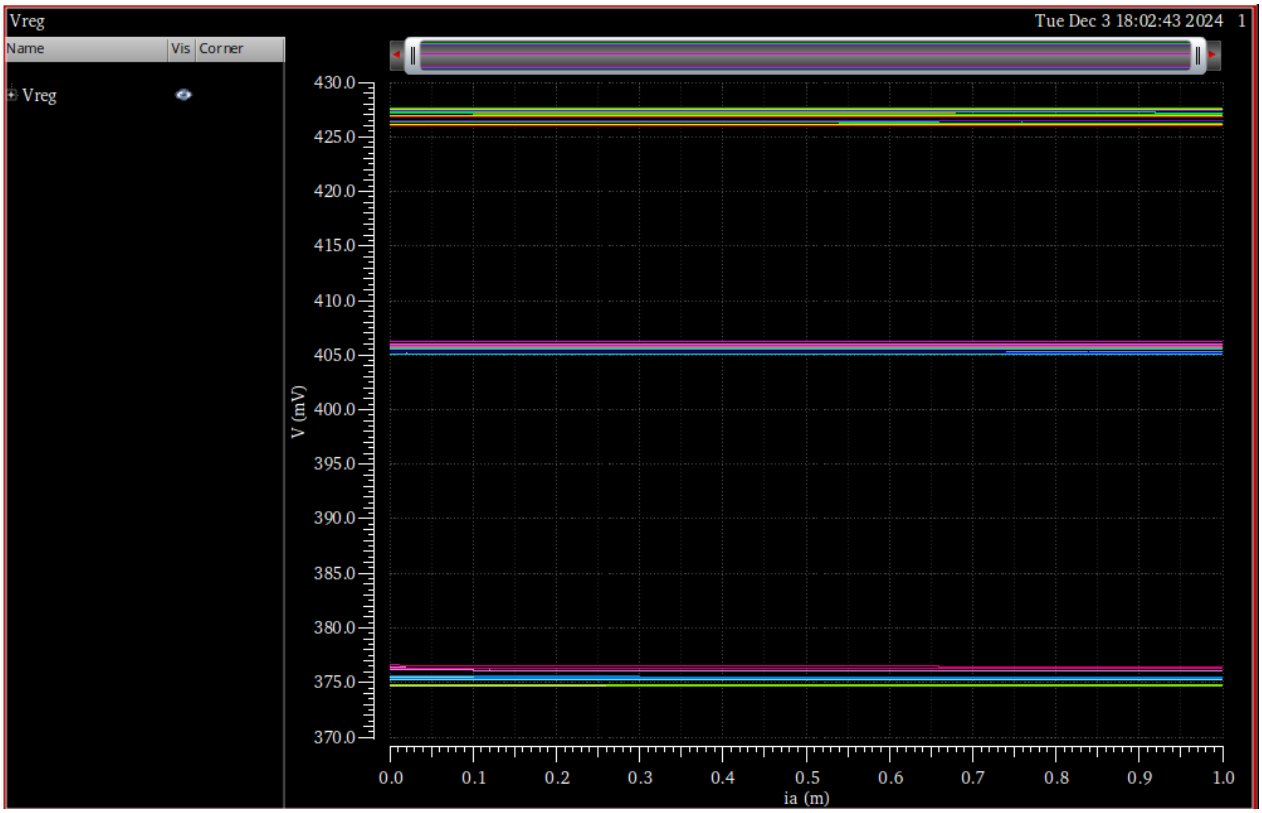
# Vreg




# Graph with varying load



# Across all corners:

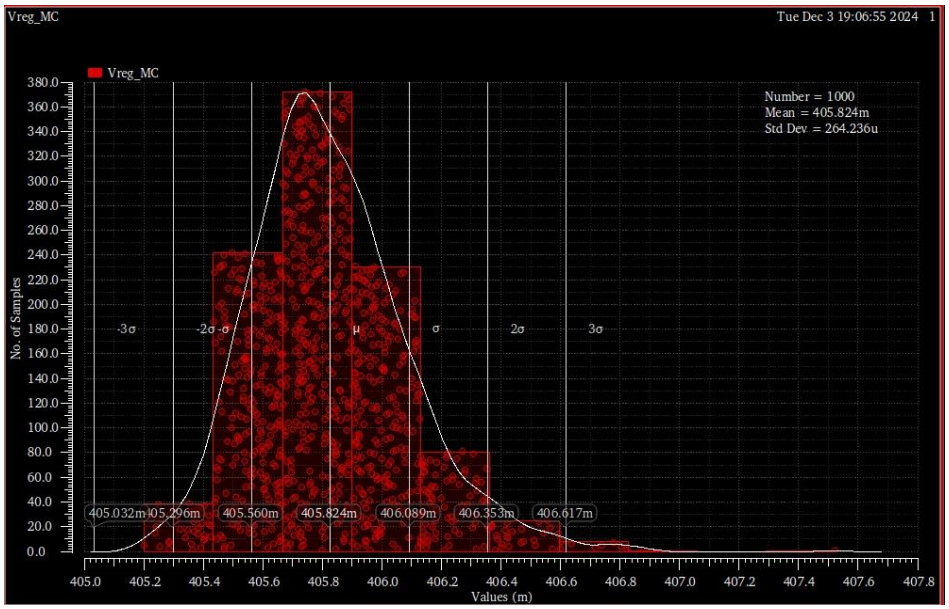



# Random variation across MC:

| Corners | Nominal | C0 | MC |
|---|---|---|---|
| **Temperature** | | -40 27 125 | 27 |
| **Design Variables** | | | |
| vdd | | 1.62 1.8 1.98 | 1.8 |
| Click to add | | | |
| **Parameters** | | | |
| Click to add | | | |
| **Model Files** | | | |
| cor_bip.scs | | tt_bip | tt_bip |
| cor_mim.scs | | tt_mim | tt_mim |
| cor_na.scs | | tt_na | tt_na |
| cor_res.scs | | tt_res | tt_res |
| cor_rfres_rpo.scs | | tt_rfres_rpo | tt_rfres_rpo |
| cor_std_mos.scs | | tt ss ff sf fs | tt ss ff sf fs |
| cr018gpii_v1d0.scs | | mc_mim | mc_mim |
| ...18gpii_v1d0.scs:1 | | stat_noise | stat_noise |
| ...18gpii_v1d0.scs:2 | | <section> | stat_res |
| ...18gpii_v1d0.scs:3 | | <section> | stat |
| ...18gpii_v1d0.scs:4 | | <section> | mc_res |
| ...18gpii_v1d0.scs:5 | | <section> | stat_mis_bip |
| ...18gpii_v1d0.scs:6 | | <section> | mc_bip |
| ...18gpii_v1d0.scs:7 | | <section> | stat_mis |
| ...18gpii_v1d0.scs:8 | | <section> | stat_mim |
| Click to add | | | |
| **Model Group(s)** | | <modelgroup> | <modelgroup> |
| Click to add | | | |
| **Tests** | | | |
| ...roj:LDO_des:1 | | | |
| **...mber of Corners** | 1 | 45 | 1 |

# No load:

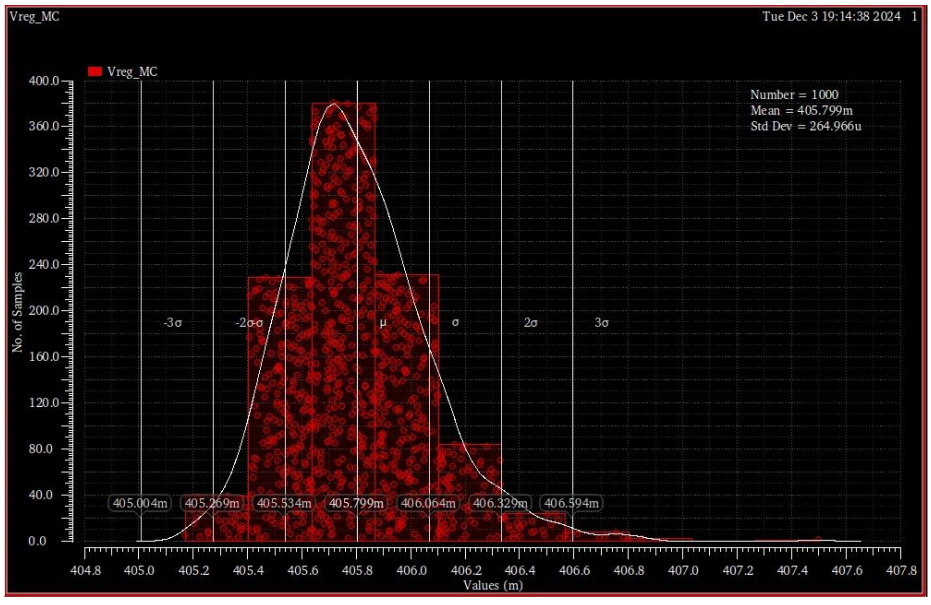
| Test | Name | Yield | Min | Target | Max | Mean | Std Dev | Cpk | Errors |
|---|---|---|---|---|---|---|---|---|---|
| Yield Estimate: 100 %(1000 passed/1000 pts) | Confidence Level:<not set> | Filter:<not set> | | | | | | | |
| Final_proj:LDO_des:1 | | | | | | | | | |
| Vreg(sum... | | 100 | 405.2m | | 407.5m | 405.8m | 264.2u | | 0 |
| Vreg_MC | | 100 | 405.2m | info | 407.5m | 405.8m | 264.2u | | 0 |




# Random variation across MC:

# 1mA load:

| Test | Name | Yield | Min | Target | Max | Mean | Std Dev | Cpk | Errors |
|---|---|---|---|---|---|---|---|---|---|
| Yield Estimate: 100 %(1000 passed/1000 pts) | Confidence Level:<not set> | Filter:<not set> | | | | | | | |
| Final_proj:LDO_des:1 | | | | | | | | | |
| Vreg(sum... | | 100 | 405.2m | | 407.5m | 405.8m | 265u | | 0 |
| Vreg_MC | | 100 | 405.2m | info | 407.5m | 405.8m | 265u | | 0 |



# Conclusion

LDO was designed and the specifications were achieved as below:

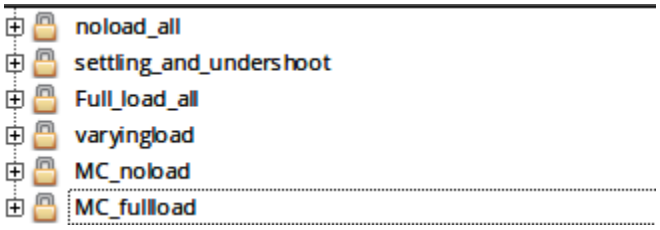
NO LOAD:

- Required Spec was less than +-10%: variation which was achieved
- UGB at Nominal 8M
- 3dB BW 55.93M
- Power Consumption less than 300u across all corners
- PSRR below 21 everywhere across all corners and PVT
- At DC PSRR is below -46 to -59
- Max undershoot at 405.66mV Undershoot value as 0.14mV
- Settling Time as 103ns

FULL LOAD:

- Less than +-10%: variation which was achieved
- UGB at Nominal 66.37M, rest all 55.33M
- 3dB BW 86M at Nominal ,rest all above 68.89M
- PSRR below 20 everywhere across all corners and PVT
- At DC PSRR is below -44 to -58

Paths:/data/proj/Low_power_design/tech_tsmc65nm/work/users/damini/Final_proj




Locked Files